CCEE Advanced Analytics Using Statistics

Session 13 & 14: Statistical Inference & Hypothesis Testing Complete Study Guide

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1. CORE TERMINOLOGY

Null Hypothesis (H_0)

Definition: Default assumption claiming no effect, no difference, or no relationship exists.

Key Concept:

- We never "accept" H_0; we either reject it or "fail to reject" it
- Burden of proof is on the alternative hypothesis
- It's the status quo: "nothing new is happening"

Examples:

- H_0: New drug has NO effect compared to old drug
- H_0: Male and female students have EQUAL average height
- H_0: Two variables are INDEPENDENT

Alternative Hypothesis (H_1 or H_a)

Definition: Competing claim that contradicts H_0; asserts that an effect or difference exists.

Directional (One-Tailed):

- H_1: $\mu > \mu_0$ (greater than)

- H_1: $\mu < \mu_0$ (less than)
- Specifies direction of effect

Non-Directional (Two-Tailed):

- H_1: $\mu \neq \mu_0$ (not equal)
- No specific direction

Choice Depends On:

- Research question specificity
- Prior knowledge or theory
- Study design

Significance Level (α) - Alpha

Definition: Maximum allowed probability of Type I error; the threshold for rejecting H_0.

Common Values:

- α = 0.05 (5%) - Most common, standard in many fields
- α = 0.01 (1%) - More stringent, used in fields where errors are costly
- α = 0.10 (10%) - More lenient, exploratory research

Critical Relationship:

- α controls Type I error (false positive rate)
- Lower α = stricter standard = less likely to falsely reject H_0
- Trade-off: Lower α increases Type II error risk

95% Confidence Interval ↔ α = 0.05 (total)

- Two-tailed: α/2 = 0.025 on each tail
- One-tailed: α = 0.05 on one side

Confidence Interval (CI)

Definition: Range of plausible values within which we believe the true population parameter lies.

Interpretation (CORRECT):

- "If we repeated this study 100 times, the true parameter would fall in our computed interval 95 times"
- NOT "95% chance the true value is in THIS specific interval"
- Focuses on long-run behavior of the method, not probability of a single interval

General Formula:
Point Estimate ± (Critical Value × Standard Error)

Example:

- 95% CI for mean height = 170 ± 2 cm = [168, 172] cm
- Interpretation: We're 95% confident true average height is between 168-172 cm

Relationship to α:

- 95% CI ↔ α = 0.05
- 99% CI ↔ α = 0.01
- 90% CI ↔ α = 0.10
- Confidence Level + Significance Level = 100%

P-Value

Definition: Probability of observing sample data as extreme (or more extreme) than what we obtained, GIVEN that H_0 is true.

Key Points:

- NOT the probability that H_0 is true
- Smaller p-value = stronger evidence against H_0
- It's a conditional probability: P(data | H_0 is true)

Decision Rule:

- If p-value ≤ α → Reject H_0 (statistically significant result)
- If p-value > α → Fail to reject H_0 (not statistically significant)

Example:

- p-value = 0.03, α = 0.05
- Since 0.03 < 0.05 → Reject H_0 → Result is statistically significant
- Interpretation: If H_0 were true, we'd see data this extreme only 3% of the time; unlikely under H_0

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2. TYPE I & TYPE II ERRORS

Error Definitions

Error Type	Definition	What Happens	Probability	Memory Aid
Type I (False Positive/α error)	Reject TRUE H_0	Claim effect exists when it DOESN'T	α (significance level)	"False alarm"
Type II (False Negative/β error)	Fail to reject FALSE H_0	Claim NO effect when it DOES exist	β	"Miss the disease"

Real-World Examples

Example 1: Medical Test

- H_0: Person is healthy
- Type I: Test says positive when person is actually healthy (false alarm)
- Type II: Test says negative when person is actually sick (dangerous miss)

Example 2: Product Quality

- H_0: Batch of products meets quality standards
- Type I: Reject good batch (waste, customer satisfaction lost)
- Type II: Accept bad batch (legal liability, reputation damage)

Example 3: Criminal Trial

- H_0: Defendant is innocent
- Type I: Convict innocent person (severe injustice)
- Type II: Acquit guilty person (criminal escapes)

Power of a Test

Definition: Probability of correctly rejecting a FALSE H_0; ability to detect effect when one truly exists.

Formula: Power = 1 - β

Example:

- If β = 0.20 (20% chance of Type II error)
- Power = 1 - 0.20 = 0.80 (80% chance of detecting effect)

Factors Affecting Power:

1. **Sample Size**: Larger n → Higher power
2. **Effect Size**: Larger true effect → Higher power
3. **Significance Level (α)**: Larger α → Higher power (but increases Type I error)
4. **Test Sensitivity**: Some tests inherently more powerful

In Practice:

- Researchers target power ≥ 0.80 (or 80%)
- This means 80% chance to detect effect if it exists
- Acceptable Type II error rate = 20%

Relationship Between Errors

Trade-off:

- Decreasing α (more strict) → Increases β
- Increasing power → May decrease α or increase sample size
- Cannot minimize both errors simultaneously without increasing n

Graphical Understanding:

- Two overlapping probability distributions (H_0 vs H_1)
- α = area in H_0 distribution beyond critical value (rejection region)
- β = area in H_1 distribution before critical value (non-rejection region)
- Shifting critical value left decreases α but increases β

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3. TAILS OF TEST

One-Tailed Test (Directional)

When Used:

- Research question specifies direction
- Prior evidence suggests effect in one direction
- Researcher specifically interested in one direction

Hypotheses:

- H_0: $\mu = \mu_0$
- H_1: $\mu > \mu_0$ (RIGHT-TAILED) OR $\mu < \mu_0$ (LEFT-TAILED)

Example:

- Q: "Does new training method INCREASE test scores?" (directional)
- H_1: $\mu_after > \mu_before$

Advantage:

- All α concentrated in one tail
- More power to detect effect in that direction
- More sensitive test in chosen direction

Disadvantage:

- Cannot detect effect in opposite direction
- Would fail to reject H_0 even if large effect exists opposite direction

Critical Value:

- For $\alpha = 0.05$ (one-tailed) in normal distribution: $z = 1.645$
- Vs. two-tailed $z = 1.96$
- One-tailed allows less extreme value because entire α on one side

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Two-Tailed Test (Non-Directional)

When Used:

- Research question asks about difference without specifying direction
- Effect could plausibly occur in either direction
- Safer, more conservative choice when unsure

Hypotheses:

- H_0: $\mu = \mu_0$
- H_1: $\mu \neq \mu_0$

Example:

- Q: "Do male and female students have DIFFERENT average heights?" (non-directional)
- H_1: μ_male ≠ μ_female

Advantage:

- Detects effects in either direction
- Safer choice when direction unknown
- Standard in most research (default)

Disadvantage:

- Less power than one-tailed test
- α split between two tails: α/2 on each side
- Requires larger effect to reach significance

Critical Value:

- For α = 0.05 (two-tailed) in normal distribution: z = 1.96
- Split: 0.025 on right tail + 0.025 on left tail
- More extreme critical value = harder to reject H_0

Decision Rule Examples

Example 1: Two-Tailed, α = 0.05, p-value = 0.03

- Compare: 0.03 ≤ 0.05? YES
- **Decision**: Reject H_0
- **Interpretation**: Statistically significant difference

Example 2: Two-Tailed, α = 0.05, p-value = 0.07

- Compare: 0.07 ≤ 0.05? NO
- **Decision**: Fail to reject H_0
- **Interpretation**: Not statistically significant (but close)

Example 3: One-Tailed (Right), α = 0.05, p-value = 0.08

- Compare: 0.08 ≤ 0.05? NO
- **Decision**: Fail to reject H_0
- Note: If this were two-tailed test, p-value comparison would still fail (0.08 > 0.05)

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4. PARAMETRIC TESTS: T-TEST

When to Use t-test

Characteristics:

- Comparing MEANS
- Continuous data (interval/ratio scale)
- Data approximately normally distributed
- Assume equal variances (for two-sample test)

Advantage Over Non-Parametric:

- More powerful when assumptions met
- Smaller sample sizes needed

Disadvantage:

- Sensitive to violations of normality, especially with small n
- Outliers can heavily influence results

One-Sample t-Test

Purpose: Compare sample mean against known population mean.

Use Cases:

- Is average student height equal to 5.8 ft?
- Does average manufacturing output equal target of 1000 units?
- Is average response time equal to industry standard?

Hypotheses:

- H_0: $\mu = \mu_0$ (sample mean equals population mean)
- H_1: $\mu \neq \mu_0$ (or >, <)

Assumptions:

1. Data normally distributed (or approximately, with $n \geq 30$)
2. Continuous data
3. Independence of observations
4. Measurement is on interval/ratio scale

Formula:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ_0 = hypothesized population mean
- s = sample standard deviation
- n = sample size

Interpretation:

- t measures how many standard errors the sample mean is from μ_0
- Larger $|t|$ → stronger evidence against H_0
- Compare to t-distribution with n-1 degrees of freedom

Example:

- Sample of 25 students, average height = 172 cm
- Population average = 170 cm

- Sample SD = 5 cm
- t = (172 - 170)/(5/√25) = 2/1 = 2.0
- With df = 24, check if p-value < 0.05

Independent Two-Sample t-Test

Purpose: Compare means of TWO independent groups.

Use Cases:

- Do male and female students have different average test scores?
- Does Group A (treatment) have different outcomes than Group B (control)?
- Are salaries different between two companies?

Hypotheses:

- H_0: $\mu_1 = \mu_2$ (group means are equal)
- H_1: $\mu_1 \neq \mu_2$ (or >, <)

Assumptions:

1. Both groups normally distributed (or large sample)
2. Observations independent within and between groups
3. Equal variances (homogeneity) in both groups
4. Continuous data

Check for Equal Variances:

- Use Levene's test before t-test
- If p-value > 0.05 → Equal variances assumed → Use standard t-test
- If p-value ≤ 0.05 → Unequal variances → Use Welch's t-test (adjustment)

Formula (Equal Variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Where:

- s_p^2 = pooled variance
- n_1, n_2 = group sample sizes

Degrees of Freedom: df = $n_1 + n_2 - 2$

Example:

- Group 1 (Males): n = 50, mean = 75
- Group 2 (Females): n = 45, mean = 72
- Pooled SD = 8
- Compare t to t-distribution to find p-value
- If p < 0.05, conclude means differ

Paired t-Test

Purpose: Compare same subjects measured at TWO different times or conditions.

Use Cases:

- Does tutoring improve test scores (same students, before-after)?
- Does meditation reduce anxiety (paired measurements)?
- Twin studies comparing one twin receiving treatment, other control
- Husband-wife comparisons

Key Characteristic:

- Data comes in PAIRS
- Eliminate individual differences → More power

Hypotheses:

- H_0: μ_d = 0 (mean difference = 0)
- H_1: μ_d ≠ 0 (or >, <)

Assumptions:

1. Differences are normally distributed
2. Pairs are independent
3. Continuous measurements
4. Random sampling

Formula:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Where:

- $\bar{d}$ = mean of differences
- s_d = standard deviation of differences
- n = number of pairs

Degrees of Freedom: df = n - 1

Example:

- 20 students take test before training (x_1) and after (x_2)
- Difference d_i = x_2_i - x_1_i for each student
- Average $\bar{d}$ = 8 points improvement
- SD of differences = 12
- t = 8/(12/√20) = 8/2.68 ≈ 3.0
- If p < 0.05, conclude training improved scores

vs. Independent t-test:

- Paired eliminates between-person variation
- More powerful for detecting effect

- Choose paired only if natural pairing exists

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5. PARAMETRIC TESTS: ANOVA

One-Way ANOVA (Analysis of Variance)

Purpose: Compare means of 3 OR MORE independent groups simultaneously.

Use Cases:

- Do students from 3 different coaching institutes have different average marks?
- Do four different fertilizers produce different crop yields?
- Does product satisfaction differ across three price points?

Why Not Multiple t-Tests?:

- Conducting multiple t-tests increases Type I error
- Each test has $\alpha = 0.05$ error rate
- Combined error rate = 1 - (1 - 0.05)^k where k = number of tests
- Example: 3 t-tests → overall $\alpha \approx 0.14$ (too high!)
- ANOVA controls overall error rate

Hypotheses:

- H_0: $\mu_1 = \mu_2 = \mu_3$ (all group means equal)
- H_1: At least one mean differs

Assumptions:

1. All groups normally distributed (or large samples)
2. Equal variances across groups (homogeneity of variance)
3. Independence of observations
4. Continuous data
5. Random sampling

Check Assumptions:

- Normality: Shapiro-Wilk test
- Equal variances: Levene's test

Key Logic:

Total Variation = Between-Group Variation + Within-Group Variation

Interpretation:

- Between-group variation = differences in group means
- Within-group variation = spread of data within each group
- If groups truly same, between ≈ within
- If groups differ, between >> within

F-Statistic Formula:

$$F = \frac{MS_{between}}{MS_{within}} = \frac{SS_{between}/(k-1)}{SS_{within}/(n-k)}$$

Where:

- MS = Mean Square (variance)
- SS = Sum of Squares
- k = number of groups
- n = total sample size

Degrees of Freedom:

- Numerator: $df_1 = k - 1$
- Denominator: $df_2 = n - k$
- Example: 3 groups, 120 total → df = (2, 117)

Interpretation:

- Large F → Groups means differ more than within-group variation
- Small F → Groups means similar to within-group variation
- Compare F to F-distribution with (k-1, n-k) df
- Get p-value from F-table or software
- If p-value ≤ α → Reject H_0 → At least one mean differs

Example Calculation:

- 3 coaching institutes, 50 students each
- Institute A: mean = 75
- Institute B: mean = 78
- Institute C: mean = 72
- ANOVA tests if these 3 means differ significantly
- Suppose F = 4.5 with df = (2, 147)
- p-value = 0.012
- Since 0.012 < 0.05 → Reject H_0 → Institutes differ

Post-Hoc Tests (After Significant ANOVA)

Important: ANOVA tells WHETHER differences exist, NOT WHERE.

Purpose of Post-Hoc: Find WHICH groups differ.

Common Methods:

- **Tukey's HSD** (Honestly Significant Difference)
- **Bonferroni Correction**
- **Scheffé Test**
- **Duncan's Multiple Range Test**

When to Use:

- Only if ANOVA is significant
- Pairwise comparisons between groups
- Control multiple comparison error

Example:

- ANOVA shows 3 institutes differ
- Tukey's test compares: A vs B, A vs C, B vs C
- Finds: A and B differ (p = 0.008), but A vs C not significant, B vs C not significant

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6. NON-PARAMETRIC TESTS: CHI-SQUARE

When to Use Non-Parametric Tests

Situations Requiring Non-Parametric:

- ✓ Data is ordinal (rankings, ratings 1-5)
- ✓ Data is categorical (counts, frequencies, proportions)
- ✓ Data is continuous but heavily skewed (not normal)
- ✓ Data has extreme outliers
- ✓ Sample size very small (n < 20) and normality unclear
- ✓ Normality assumption clearly violated

Advantages:

- No normality assumption
- Work with ordinal data
- Robust to outliers
- Simpler, don't require parameters like mean/SD

Disadvantages:

- Less power than parametric (when parametric assumptions met)
- More difficult to compute (often rank-based)
- Don't test specific parameters (mean, variance)

Chi-Square (χ^2) Test of Independence

Purpose: Test association between TWO categorical variables.

Use Cases:

- Is gender (Male/Female) associated with product preference (A/B/C)?
- Does smoking status relate to lung disease occurrence?
- Do education level and income level correlate?

Hypotheses:

- H_0: Two variables are INDEPENDENT (no association)
- H_1: Two variables are ASSOCIATED (dependent)

Data Format: Contingency table (cross-tabulation)

Example Table: Gender vs Product Preference

Gender	Product A	Product B	Total
Male	60	40	100
Female	50	50	100
Total	110	90	200

Formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where:

- O = Observed count in cell
- E = Expected count under H_0 (if independent)

Calculate Expected Counts:

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Example:

- Expected for (Male, Product A) = (100 × 110)/200 = 55
- Expected for (Male, Product B) = (100 × 90)/200 = 45
- Expected for (Female, Product A) = (100 × 110)/200 = 55
- Expected for (Female, Product B) = (100 × 90)/200 = 45

Compute χ^2:

$$\chi^2 = \frac{(60 - 55)^2}{55} + \frac{(40 - 45)^2}{45} + \frac{(50 - 55)^2}{55} + \frac{(50 - 45)^2}{45}$$

$$= \frac{25}{55} + \frac{25}{45} + \frac{25}{55} + \frac{25}{45} \approx 1.2$$

Degrees of Freedom:

$$df = (rows - 1) \times (columns - 1)$$

Example: df = (2-1) × (2-1) = 1

Assumptions:

1. Independence of observations (each cell represents independent counts)
2. Expected count ≥ 5 in each cell
3. Random sampling
4. Categorical data

If Expected < 5:

- Combine categories

- Use Fisher's exact test (for 2×2 tables)
- Acknowledge limitation

Decision:

- If p-value ≤ α → Reject H_0 → Variables associated
- If p-value > α → Fail to reject H_0 → No association

Interpretation Example:

- $\chi^2 \approx 1.2$ with df = 1
- p-value ≈ 0.27
- Since 0.27 > 0.05 → Fail to reject H_0
- Conclusion: Gender and product preference are independent (not associated)

Chi-Square Goodness-of-Fit Test

Purpose: Test if observed frequencies match expected distribution.

Use Cases:

- Is a die fair? (Each face should appear 1/6 of rolls)
- Do customers prefer products A, B, C equally? (Each 1/3)
- Does population match expected demographic distribution?

Hypotheses:

- H_0: Observed frequencies match expected distribution
- H_1: Observed frequencies differ from expected

Formula: Same as above

Example: Die Fairness

- Roll die 60 times
- Expected: Each face 10 times (60/6)
- Observed: 1→12, 2→8, 3→10, 4→14, 5→9, 6→7

Compute χ^2:

$$\chi^2 = \frac{(12-10)^2}{10} + \frac{(8-10)^2}{10} + \ldots = \frac{4}{10} + \frac{4}{10} + 0 + \frac{16}{10} + \frac{1}{10} + \frac{9}{10} = 3.4$$

Degrees of Freedom:

$$df = k - 1 - p$$

Where:

- k = number of categories
- p = number of parameters estimated from data

Example: df = 6 - 1 = 5 (no parameters estimated)

Decision:

- If p-value ≤ α → Data doesn't fit expected distribution
- If p-value > α → Data consistent with expected distribution

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7. NON-PARAMETRIC TESTS: MANN-WHITNEY U-TEST

When to Use Mann-Whitney U

Purpose: Compare TWO independent groups; alternative to independent t-test.

Use Cases:

- Satisfaction ratings (1-5) differ between App A and App B?
- Test scores differ between untreated and treated groups (non-normal data)?
- Salary ranges (ordinal: low/medium/high) differ by gender?

Advantages Over t-Test:

- No normality assumption
- Works with ordinal data
- Robust to outliers
- Smaller sample sizes acceptable

Disadvantages:

- Less power than t-test when data normal
- Complex computation
- Tests entire distribution, not just location

Mann-Whitney U Test Logic

Key Idea: Rank-Based

Steps:

1. Combine data from both groups
2. Rank all observations from smallest (1) to largest ($n_1 + n_2$)
3. Sum ranks for Group 1: R_1
4. Sum ranks for Group 2: R_2
5. Compute U statistic
6. Compare to Mann-Whitney distribution

Formula:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

$$U = \min(U_1, U_2)$$

Interpretation:

- U measures difference in rank distributions
- Small U → Groups rank very differently → Strong evidence of difference
- Large U → Groups rank similarly → No difference
- Extreme U values lead to small p-values

Mann-Whitney U Example

Scenario: Two apps' satisfaction ratings (1-5)

App A data: 3, 4, 4, 5
App B data: 1, 2, 3, 3

Combined ranked: 1(B), 2(B), 3(A), 3(B), 4(A), 4(A), 5(A)

- Ranks 1-7

When values tied (same score):

- Assign average rank
- Example: 3 appears at positions 3 and 4 → Both get rank 3.5

Recalculate:
Combined ranked: 1(B), 2(B), 3.5(A), 3.5(B), 5(A), 6(A), 7(A)

Sum ranks:

- R_A = 3.5 + 5 + 6 + 7 = 21.5
- R_B = 1 + 2 + 3.5 = 6.5
- Check: 21.5 + 6.5 = 28 = 7(8)/2 ✓

Compute U:

$$U_1 = 4 \times 4 + \frac{4 \times 5}{2} - 21.5 = 16 + 10 - 21.5 = 4.5$$

$$U_2 = 4 \times 4 + \frac{4 \times 5}{2} - 6.5 = 16 + 10 - 6.5 = 19.5$$

$$U = \min(4.5, 19.5) = 4.5$$

Lookup p-value: For $n_1 = n_2 = 4$, U = 4.5, $p \approx 0.08$

- If $\alpha = 0.05$ and $p = 0.08 > 0.05$ → Fail to reject H_0

Hypotheses for Mann-Whitney U

H_0: Two groups have same distribution (medians approximately equal)
H_1: Two groups have different distributions

Note: Technically tests distributions, not just location (median).

Assumptions:

1. Independence of observations
2. Ordinal or continuous data

3. Random sampling
4. Similar distribution shapes (optional but strengthens interpretation)

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8. HYPOTHESIS TESTING WORKFLOW (Generic Steps)

The Seven-Step Process

Step 1: Formulate Hypotheses
├─ State H_0 (null: no effect/no difference)
└─ State H_1 (alternative: specific effect)
↓
Step 2: State Significance Level (α) & Decide Tails
├─ Choose α (typically 0.05)
└─ Decide one-tailed or two-tailed
↓
Step 3: Choose Appropriate Test
├─ Ask: How many groups? What data type? Normal?
└─ Select: t-test, ANOVA, chi-square, Mann-Whitney U, etc.
↓
Step 4: Calculate Test Statistic from Sample Data
├─ Use formula for chosen test
├─ Gather necessary statistics (means, SD, frequencies, etc.)
└─ Compute t, F, χ^2, U, or other statistic
↓
Step 5: Find P-Value
├─ Compare statistic to reference distribution
├─ Use tables, software, or calculator
└─ Determine probability of observing this extreme data under H_0
↓
Step 6: Decision
├─ IF p-value ≤ α:
│ └─ REJECT H_0 (accept H_1)
│ └─ Result is STATISTICALLY SIGNIFICANT
└─ IF p-value > α:
└─ FAIL TO REJECT H_0
└─ Insufficient evidence for effect
↓
Step 7: Conclusion (in context of research question)
├─ Interpret what result means practically
├─ Discuss implications
└─ Note limitations and next steps

Example: Full Hypothesis Test Walkthrough

Research Question: Does a new tutoring program improve test scores?

Step 1: Formulate Hypotheses

- H_0: μ_after = μ_before (tutoring has no effect)
- H_1: μ_after > μ_before (tutoring increases scores) [ONE-TAILED]

Step 2: Significance Level & Tails

- $\alpha = 0.05$ (standard level)
- One-tailed (directional: specifically increase)

Step 3: Choose Test

- Same students measured twice → Paired t-test
- Check data: approximately normal, no extreme outliers ✓

Step 4: Calculate Test Statistic

- 25 students, average improvement $\bar{d} = 5$ points
- SD of differences $s_d = 8$ points
- $t = 5/(8/\sqrt{25}) = 5/1.6 = 3.125$

Step 5: Find P-Value

- $df = 25 - 1 = 24$
- $t = 3.125$, $df = 24$
- Using t-table or software: $p \approx 0.002$ (one-tailed)

Step 6: Decision

- Is $p \leq \alpha$? Is $0.002 \leq 0.05$? YES
- **REJECT H_0**

Step 7: Conclusion

- We have statistically significant evidence ($p = 0.002$) that the tutoring program improves test scores
- On average, scores improved by 5 points (95% CI: 2.7 to 7.3 points)
- This improvement is unlikely due to chance alone

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9. TEST SELECTION DECISION TREE

How Many Groups Are You Comparing?

ONE GROUP
├─ Comparing sample mean to known population value?
└─ → One-Sample t-Test (if continuous, normal)
→ Chi-Square Goodness-of-Fit (if categorical)
↓
TWO GROUPS
├─ Data Type?
│ ├─ Continuous (interval/ratio)
│ │ ├─ Both groups normal & independent?
│ │ │ ├─ YES → Independent t-Test
│ │ │ └─ NO (skewed/outliers) → Mann-Whitney U
│ │ └─ Same subjects measured twice?
│ │ ├─ YES → Paired t-Test
│ │ └─ NO → Independent t-Test
│ └─ Categorical (counts/frequencies)

```
│ └─ → Chi-Square Test of Independence
↓
THREE OR MORE GROUPS
├─ Data Type?
│ ├─ Continuous (interval/ratio)
│ │ ├─ All groups normal?
│ │ │ ├─ YES → One-Way ANOVA
│ │ │ └─ NO (skewed/outliers) → Kruskal-Wallis Test
│ │ └─ Repeated measures (same subjects)?
│ │ ├─ YES → Repeated Measures ANOVA
│ │ └─ NO → One-Way ANOVA
│ └─ Categorical (counts/frequencies)
│ └─ → Chi-Square Test (multiple categories)
```

Quick Selector Checklist

Before Selecting Test, Ask:

- [ ] How many groups? (1, 2, 3+)
- [ ] What scale is data? (Categorical, Ordinal, Continuous)
- [ ] Is data normally distributed? (Check histogram, Q-Q plot, Shapiro-Wilk)
- [ ] Are variances equal? (Check Levene's test)
- [ ] Are observations independent? (Design question)
- [ ] Sample size? (Large n ≥ 30 more forgiving of non-normality)
- [ ] Any extreme outliers? (May suggest non-parametric)

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10. QUICK COMPARISON TABLE

Feature	One-Sample t	Independent t	Paired t	ANOVA	Chi-Square	Mann-Whitney U
Groups	1	2	2	3+	2+	2
Comparison	vs known value	Between groups	Within (before-after)	Between groups	Association	Between groups
Data Type	Continuous	Continuous	Continuous (differences)	Continuous	Categorical	Ordinal/Continuous
Assumption	Normal	Normal + Equal var	Differences normal	Normal + Equal var	None	No normality
Basis	Means	Means	Mean differences	Means	Frequencies	Ranks/Distribution
Parametric?	Yes	Yes	Yes	Yes	No	No
Test Statistic	t	t	t	F	χ^2	U
Power	High (if assumptions)	High (if assumptions)	High	High (if assumptions)	Low-Medium	Medium-High

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11. FORMULAS AT A GLANCE

Test Statistic Formulas

One-Sample t-Test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Independent Two-Sample t-Test (equal variances):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}}$$

Where pooled variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Paired t-Test:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

ANOVA F-Statistic:

$$F = \frac{MS_{between}}{MS_{within}} = \frac{SS_{between}/(k-1)}{SS_{within}/(n-k)}$$

Chi-Square:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Mann-Whitney U:

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Confidence Interval (General Form)

$$\text{CI} = \text{Point Estimate} \pm (\text{Critical Value} \times \text{Standard Error})$$

For mean (t-based):

$$\text{CI} = \bar{x} \pm t_{crit} \times \frac{s}{\sqrt{n}}$$

Relationships to Remember

- **Confidence Level + Significance Level = 100%**
  - 95% CI ↔ α = 0.05
  - 99% CI ↔ α = 0.01
  - 90% CI ↔ α = 0.10
- **Power and Error**
  - Power = 1 - β
  - Type I error = α
  - Type II error = β

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12. COMMON MCQ PATTERNS & TRAPS

Trap 1: Type I vs Type II Error

✖ **INCORRECT**:

- "Type I error is failing to reject false H_0"
- "Type II error is rejecting true H_0"

✓ **CORRECT**:

- Type I (False Positive) = Rejecting TRUE H_0 = Probability α
- Type II (False Negative) = Failing to reject FALSE H_0 = Probability β

Memory: Think "Type I is like condemning innocent" vs "Type II is like letting guilty go free"

Trap 2: One-Tailed vs Two-Tailed

✖ **INCORRECT**: Always assume two-tailed

✓ **CORRECT**: Read question carefully

- "Different/change/vary" → Two-tailed
- "Greater than / Less than / Better / Improved" → One-tailed
- One-tailed is more specific, two-tailed is safer

Math:

- One-tailed $\alpha = 0.05$ uses all in one direction
- Two-tailed $\alpha = 0.05$ splits to $\alpha/2 = 0.025$ each side
- One-tailed more powerful in chosen direction

Trap 3: P-Value Interpretation

✖ **INCORRECT**:

- "p-value is probability that H_0 is true"
- "p-value is probability of Type I error"
- "p-value > 0.05 means H_0 is true"

✓ **CORRECT**:

- p-value = P(data this extreme | H_0 is true)
- p-value $\leq \alpha$ → Reject H_0 (evidence against it)
- p-value > α → Fail to reject H_0 (insufficient evidence, not proof of truth)

Trap 4: Parametric vs Non-Parametric

✖ **INCORRECT**:

- "Always use parametric tests (more powerful)"
- "Non-parametric tests don't work with continuous data"

✓ **CORRECT**:

- Check assumptions first
- Parametric: More powerful IF assumptions met
- Non-parametric: Use if assumptions violated, works with ordinal data
- When unsure and n ≥ 30, parametric usually safe

Trap 5: Test Selection

✖ **INCORRECT**: "I need to compare groups" → Automatically pick t-test

✓ **CORRECT**: Ask systematically

- How many groups? (1, 2, 3+)
- Continuous or categorical data?
- Normal distribution?
- Independent or paired?
- Then choose test

Trap 6: ANOVA Result Interpretation

✖ **INCORRECT**:

- "ANOVA tells which groups differ"
- "ANOVA tests specific group differences"

✓ **CORRECT**:

- ANOVA answers: "Does at least one mean differ?" (YES or NO)
- To find WHICH groups differ → Use Post-hoc tests (Tukey, Bonferroni)

Trap 7: CI and Alpha Relationship

✖ **INCORRECT**:

- "95% CI means $\alpha = 0.95$"
- "99% CI means $\alpha = 0.99$"

✓ **CORRECT**:

- 95% CI ↔ $\alpha = 0.05$ (inverse relationship)
- 99% CI ↔ $\alpha = 0.01$
- Confidence + Significance = 100%

Trap 8: Chi-Square Expected Counts

✖ **INCORRECT**:

- "All expected counts must be ≥ 10"
- "Chi-Square cannot be used if any expected < 5"

✓ **CORRECT**:

- Expected counts should be ≥ 5 (in most cells)
- If violated: combine categories, use Fisher's exact test, or note limitation
- Very small expected counts make test unreliable

Trap 9: Mann-Whitney U Interpretation

✖ **INCORRECT**:

- "Mann-Whitney U tests if medians are equal"
- "Mann-Whitney U is just a 'non-parametric t-test'"

✓ **CORRECT**:

- Mann-Whitney U tests if distributions are same (including location, shape, spread)
- Primarily about rank distributions, not specifically medians
- Use when normality violated or data ordinal

Trap 10: Sample Size in Hypothesis Testing

✖ **INCORRECT**:

- "Larger sample size always gives larger p-value"
- "Small p-value always means large effect size"

✓ **CORRECT**:

- Larger n → More power to detect effect → Smaller p-values for same effect
- Small p-value means statistically significant, NOT necessarily practically important
- Use effect size (Cohen's d, etc.) to assess practical significance

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13. REAL-WORLD ANALOGIES

Hypothesis Testing as Court Trial

Mapping:

Concept	Court	Statistics
Default position	"Innocent until proven guilty"	H_0 (no effect)
Burden of proof	Prosecution	Alternative hypothesis (H_1)
Evidence	Witness testimony, physical evidence	Sample data
Judge's decision threshold	"Beyond reasonable doubt"	α (significance level)
Strength of evidence	How compelling testimony is	p-value
Wrong verdict 1	Convicting innocent person	Type I error (false positive)
Wrong verdict 2	Acquitting guilty person	Type II error (false negative)
Appeal/review	Case reviewed; verdict upheld or overturned	Replication studies; meta-analysis

Decision Process:

- Prosecution (H_1) presents evidence (p-value)
- Judge has standard: "Beyond reasonable doubt" (α)
- If evidence strong enough ($p \leq \alpha$): Convict (reject H_0)
- If evidence weak ($p > \alpha$): Acquit (fail to reject H_0)

Type I vs Type II Errors (Medical Example)

Scenario: Testing person for rare disease

Type I Error (False Positive):

- Test says: "You have disease"
- Reality: "You don't have disease"
- Consequence: Unnecessary anxiety, possibly unnecessary treatment/surgery
- Cost: Emotional, financial, medical risk from treatment

Type II Error (False Negative):

- Test says: "You don't have disease"
- Reality: "You do have disease"
- Consequence: Missing diagnosis, delayed treatment

- Cost: Disease progresses, potentially fatal

Which is worse?

- Depends on disease severity and treatment side effects
- Rare disease with mild treatment → Type II worse (miss diagnosis)
- Common disease with harsh treatment → Type I worse (unnecessary treatment)

Parametric vs Non-Parametric (Class Scores Example)

Parametric Approach (t-test):

- "Class A average = 75 marks, Class B average = 72 marks"
- Compare the CENTER (mean) directly
- Assumes: Scores normally distributed, no extreme outliers
- More powerful if assumptions hold

Non-Parametric Approach (Mann-Whitney U):

- Rank all 100 students from lowest to highest score
- Class A students tend to rank higher → Evidence of difference
- Works even if scores skewed, ordinal, or outliers
- More robust; less power if assumptions were met

Analogy:

- Parametric = Compare average salaries directly
- Non-parametric = Compare median job positions in ranked list (positions: junior, middle, senior)

ANOVA vs Repeated t-tests (Time Savings Analogy)

Why Not Compare 3 Groups with Three t-tests?

Imagine: Homework graded by 3 teachers, concerned about fairness.

Wrong Way: Compare Teacher A vs B, A vs C, B vs C (three separate tests)

- Each test has 5% error rate ($\alpha = 0.05$)
- Combined error rate ≈ 14% (too high!)
- Like running red light three times → Getting caught becomes much more likely

Right Way: ANOVA (one overall test)

- Keeps total error at 5%
- More efficient; avoids multiple testing problem
- Like going through one intersection carefully → Much safer

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14. PRACTICE MCQS & SOLUTIONS

Practice Set 1: Terminology & Concepts

Q1: In hypothesis testing, the p-value represents:
A. Probability that the null hypothesis is true
B. Probability of observing data as extreme as obtained, given H_0 is true
C. Probability of Type I error
D. Significance level chosen before test

Answer: B
Explanation: p-value is specifically P(data | H_0), not the probability H_0 is true. Option C (probability of Type I error = α) is chosen before test, not calculated.

Q2: Which statement about 95% confidence interval is CORRECT?
A. 95% chance true parameter is in this specific interval
B. This method captures true parameter 95% of the time in repeated sampling
C. 95% of data falls within interval
D. 5% confidence that H_0 is true

Answer: B
Explanation: CI refers to long-run behavior of the method, not probability of a single interval. Option A is common misinterpretation.

Q3: Type II error and power are related by:
A. Power + β = 1
B. Type II error = α
C. Power = 1 - β
D. Power = β

Answer: C
Explanation: Power represents ability to detect true effect; β is probability of missing it. They sum to 1.

Q4: A two-tailed test with α = 0.05 has critical value ±1.96 for normal distribution because:
A. 0.05/2 = 0.025 on each tail
B. 0.05 means 95% confidence
C. Total area in both tails = 0.05
D. All above

Answer: D
Explanation: All statements are accurate. The 0.05 is split between two tails.

Practice Set 2: Test Selection

Q5: Researcher wants to compare average anxiety scores among 4 different stress-management techniques. What is most appropriate?
A. Four independent t-tests
B. One-Way ANOVA
C. Chi-Square test
D. Mann-Whitney U-test

Answer: B
Explanation:

- 4 groups (3+ triggers ANOVA, not t-test)
- Comparing means (ANOVA, not Chi-Square)
- Continuous data, not ordinal (not Mann-Whitney U)
- Using one ANOVA avoids multiple testing problem

Q6: Customer satisfaction ratings (1-5 scale) are heavily skewed. Comparing two groups, which test is best?
A. Independent t-test
B. Paired t-test
C. Mann-Whitney U-test
D. Chi-Square test

Answer: C
Explanation:

- Ordinal data (1-5 ratings) suggests non-parametric
- Skewed distribution violates t-test normality assumption
- Mann-Whitney U works with ordinal, non-normal data
- Not Chi-Square (not testing categorical association)

Q7: Same participants tested before and after meditation program. Test scores improved on average. Which test?
A. Independent t-test
B. Paired t-test
C. One-Way ANOVA
D. Mann-Whitney U-test

Answer: B
Explanation:

- Same subjects measured twice → Paired t-test (not independent t)
- Two time points (before/after), not 3+ groups
- Continuous data, not categorical
- Paired design eliminates individual differences

Q8: Gender (M/F) vs Brand Preference (A/B/C) association. Best test?
A. t-test
B. ANOVA
C. Chi-Square test
D. Mann-Whitney U-test

Answer: C
Explanation:

- Both variables categorical
- Testing association/independence
- Chi-Square designed for categorical data
- Create contingency table; test independence

Practice Set 3: Interpretation & Decisions

Q9: t-test yields t = 2.5, df = 48, two-tailed test, α = 0.05. p-value ≈ 0.015. Decision?
A. Reject H_0; statistically significant
B. Fail to reject H_0; not significant
C. Increase α to make it significant
D. Conduct additional tests

Answer: A
Explanation:

- p-value = 0.015 < α = 0.05
- Decision rule: $p \leq \alpha \rightarrow$ Reject H_0
- Result is statistically significant
- Do not adjust α after seeing results (invalid)

Q10: ANOVA F = 1.2, df = (3, 96), p-value = 0.31, α = 0.05. Conclusion?
A. At least one group mean differs significantly
B. All group means are equal
C. Insufficient evidence to conclude means differ
D. Type II error has occurred

Answer: C
Explanation:

- p = 0.31 > α = 0.05
- Fail to reject H_0
- Cannot conclude means differ based on this data
- This does NOT prove means are equal; just insufficient evidence
- Option D is incorrect (Type II error is probability, not event)

Q11: Chi-Square test for independence yields $\chi^2 = 0.8$, df = 1, p-value = 0.37. What's the relationship between variables?
A. Strong association (p < 0.05)
B. Weak association (p > 0.05)
C. No association detectable in data
D. Chi-Square is invalid for this data

Answer: C
Explanation:

- p = 0.37 > α = 0.05
- Fail to reject H_0 of independence
- No statistically significant association detected
- Does NOT prove no relationship; just no evidence in this sample

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Q12: Before running t-test, Levene's test p-value = 0.02. Decision?
A. Use standard t-test (equal variances)
B. Use Welch's t-test (unequal variances)
C. Don't run t-test; use Mann-Whitney U
D. Ignore Levene's test; run standard t-test anyway

Answer: B
Explanation:

- Levene's test $p = 0.02 < 0.05$
- Reject null of equal variances
- Use Welch's t-test (adjustment for unequal variances)
- Still better to use t-test (option C unnecessary unless other violations)

Practice Set 4: Error Types & Power

Q13: In clinical trial, Type II error means:
A. Declaring effective drug ineffective
B. Declaring ineffective drug effective
C. Using wrong statistical test
D. Calculating p-value incorrectly

Answer: A
Explanation:

- Type II = Fail to reject false H_0
- H_0: Drug has no effect
- Type II: Conclude drug ineffective (fail to reject) when it actually is effective
- This is missing real effect (dangerous in medicine)

Q14: To increase statistical power:
A. Decrease α
B. Decrease sample size
C. Increase sample size
D. Use one-tailed test instead of two-tailed

Answer: C and D (check if multiple answers allowed)
Explanation:

- Larger sample → More power (more data)
- One-tailed → All α in one direction → More power in that direction
- Decreasing α decreases power (trade-off)
- Decreasing n decreases power

If only one answer: **C** is most reliably increases power
If multiple: Both C and D correct

Q15: $\alpha = 0.05$ and $\beta = 0.20$. What is power and what is Type I error rate?
A. Power = 0.05, Type I = 0.20
B. Power = 0.80, Type I = 0.05
C. Power = 0.20, Type I = 0.80
D. Power = 0.95, Type I = 0.20

Answer: B
Explanation:

- Power = $1 - \beta = 1 - 0.20 = 0.80$
- Type I error rate = $\alpha = 0.05$

- 80% chance to detect effect, 5% false alarm rate

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15. EXAM STRATEGY TIPS FOR CCEE

Before Test

✓ **Memorize Core Definitions**

- Type I = Reject TRUE H_0
- Type II = Fail to reject FALSE H_0
- p-value = P(data | H_0)
- α = maximum allowed Type I error

✓ **Practice Test Selection Systematically**

- Draw decision tree on scratch paper
- Ask: How many groups? What data type?
- Practice until automatic

✓ **Know When to Use Parametric vs Non-Parametric**

- Parametric: Normal, continuous, higher power
- Non-parametric: Non-normal, ordinal, robust

✓ **Understand One-Tailed vs Two-Tailed**

- Two-tailed: "Different" (safer default)
- One-tailed: "Greater/Less" (specific direction)

✓ **Master Formulas and Degrees of Freedom**

- One-sample t: df = n - 1
- Two-sample t: df = $n_1 + n_2 - 2$
- ANOVA: df = (k-1, n-k)
- Chi-Square: df = (rows-1)(columns-1)

During MCQ Exam

1. Read Carefully

- Identify: How many groups? Continuous or categorical?
- Check: Are assumptions mentioned? (normal, equal variances, etc.)

2. Use Process of Elimination

- Eliminate obviously wrong answers
- Example: "3 groups" + "means" → Not t-test

3. Watch for Common Traps

- Type I vs II confusion
- One-tailed vs two-tailed wording
- p-value interpretation
- ANOVA post-hoc requirement

4. Decision Rule Anchor

- Remember: $p \leq \alpha \rightarrow$ Reject H_0
- Everything else follows from this

5. Time Management

- Don't overthink; use flowchart
- Mark difficult; return later
- Aim for 1-1.5 min per question

6. Quick Elimination Rules

- 3+ groups + means → ANOVA (not t)
- Categorical data → Chi-Square
- Ordinal/skewed data → Mann-Whitney U or non-parametric
- Same subjects twice → Paired t-test

Scoring Strategies

High Confidence (Answer Immediately):

- Clear wording, straightforward test selection
- Direct formula application
- Unambiguous interpretation

Medium Confidence (Re-Read, Decide):

- Multiple similar-seeming options
- Complex scenario
- Requires careful distinction (parametric vs non)

Low Confidence (Educated Guess or Skip):

- Thoroughly confused
- Guess based on best logic
- Mark for review if time

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16. KEY TAKEAWAYS FOR CCEE

The "Must-Know" Essentials

1. **Type I error** (False Positive) = Reject TRUE H_0 = Probability α
2. **Type II error** (False Negative) = Fail to reject FALSE H_0 = Probability β
3. **Power** = **1 - β** = Probability of correctly detecting effect
4. **Parametric Tests** (t, ANOVA) = Assume normality, mean-based, more powerful
5. **Non-Parametric Tests** (Chi-Square, U) = No normality, rank/count-based, robust
6. **Test Selection** = Ask: Groups? Data type? Normal?
    - 1 group vs value → One-sample t
    - 2 groups, normal → t-test (independent or paired)
    - 2 groups, non-normal → Mann-Whitney U
    - 3+ groups, normal → ANOVA
    - Categorical association → Chi-Square

7. **ANOVA** (3+ groups) → F = MSbetween/MSwithin → Tests IF difference exists (not WHERE)
8. **Chi-Square** (categorical) → $\chi^2 = \Sigma(O-E)^2/E$ → Tests association/independence
9. **Mann-Whitney U** (2 groups, ordinal/non-normal) → Rank-based, no normality needed
10. **p-value ≤ α** → Reject H_0 → Statistically significant
    p-value > α → Fail to reject H_0 → Not significant (not proven false)
11. **95% CI ↔ α = 0.05**; **99% CI ↔ α = 0.01** (Confidence + Significance = 100%)
12. **One-Tailed** = "Greater/Less" = More power in one direction
    Two-Tailed = "Different" = Can detect either direction (standard)
13. **Post-Hoc Tests** (Tukey, Bonferroni) = Run AFTER significant ANOVA to find WHICH groups differ
14. **Expected Counts ≥ 5** = Chi-Square assumption; if violated, combine or use alternative
15. **Effect Size ≠ Statistical Significance** = Large p is not proof of no effect (might just have small n)

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17. REFERENCE SECTION & QUICK LOOKUP

When p-value = 0.03, α = 0.05 (two-tailed)?

✓ Since 0.03 < 0.05 → Reject H_0 → Statistically significant result

When to Use Paired t-test?

✓ Same subjects measured twice (before-after)
✓ Matched pairs (twins, husband-wife)
✓ Repeated measures experimental design

✗ DO NOT use for completely independent groups

ANOVA Tells You What?

✓ WHETHER at least one group mean differs significantly
✓ IF the overall F-statistic is significant

✗ DOES NOT tell which specific groups differ (need post-hoc)

Chi-Square Expected Count Rule?

✓ Expected count ≥ 5 in each cell (recommendation)
✗ If violated: Combine categories or use Fisher's exact test

Mann-Whitney U When?

✓ Two independent groups
✓ Ordinal data (ratings, ranks)
✓ Continuous but non-normal (skewed, outliers)

✗ Three or more groups → Use Kruskal-Wallis

Degrees of Freedom Quick Reference

Test	Formula	Example
One-sample t	$n - 1$	$n=25 \rightarrow df=24$
Two-sample t	$n_1 + n_2 - 2$	$n_1=30, n_2=35 \rightarrow df=63$
Paired t	$n - 1$	$n=20$ pairs $\rightarrow df=19$
ANOVA	$(k-1, n-k)$	$k=4$ groups, $n=100 \rightarrow df=(3,96)$
Chi-Square	$(r-1)(c-1)$	3 rows, 2 cols $\rightarrow df=2$

This comprehensive study guide covers Sessions 13 & 14 of the Advanced Analytics using Statistics module for CCEE preparation. Master these concepts and decision-making frameworks to excel in the exam.

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